

Synergistic Promotion of Single-Atom Co Surrounding a PtCo Alloy Based on a g-C₃N₄ Nanosheet for Overall Water Splitting

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1 **1. Characterization**

2 Inductively coupled plasma-optical emission spectrometry (ICP-OES) was conducted
3 using a Horiba Ultra 2 instrument equipped with photomultiplier tube detection. The
4 detailed crystal and chemical structures of the synthesized photocatalyst samples were
5 evaluated by X-ray diffraction (XRD; Bruker D8 Advance, Cu target, 20 kV for voltage
6 and 2 mA for current, 5°/min for scanning speed from 10° to 80°), and Fourier transform
7 infrared (FTIR) spectroscopy (Ledon FTIR-650). X-ray photoelectron spectroscopy
8 (XPS) was conducted in a Physical Electronics Instruments Quantum 2000
9 spectrometer using monochromatic Al K α radiation generated from an electron beam
10 operated at 15 kV and 32.3 W. The spectra were collected under ultra-high vacuum
11 conditions (5×10^{-8} Pa) at a pass energy of 46.95 eV. All spectra were referenced to the
12 C 1s peak at 288.1 eV. HAADF-STEM measurements were conducted with a JEOL
13 JEM ARM 200CF aberration-corrected scanning transmission electron microscope at
14 200 kV accelerating voltage. XAFS measurements of the samples were carried out in
15 Shanghai Synchrotron Radiation Facility (SSRF). Transient photocurrent measurement:
16 First, clean the original FTO glasses with ethanol for 0.5 h ultrasonically. A 2.0 mg
17 catalyst powder was dispersed in 500 μ L terpineol and then treated with ultrasound for
18 1 h. Subsequently, the resulting suspension was spun onto a 1 cm $\times$ 1 cm FTO glass.
19 The glass was further dried in a vacuum at 80 °C for 6 h. Afterward, a CHI 660E
20 electrochemical workstation was used to carry out the photoelectrochemical tests. A
21 standard three-electrode system was used with 0.2 M Na₂SO₄ as the electrolyte. The
22 photoelectrode, counter electrode, and reference electrode were FTO glass, Pt foil, and
23 Ag/AgCl, respectively. Photoluminescence (PL) emission spectra were obtained using
24 a fluorescence spectrophotometer (Shimadzu RF-5301). Take 2mg sample powder for
25 sample preparation, put the sample into the sample tank, and pave it with a compaction
26 thickness of 1mm. PL lifetime decays were measured by an inverted-type scanning
27 confocal microscope (MicroTime-200, Picoquant). A single-mode pulsed diode laser
28 (excitation wavelength at 375 nm with an instrumental response function of $\sim$ 240 ps in
29 pulse width) was used to as an excitation source to excite the samples coated on glass
30 substrate. A dichroic mirror (490 DCXR, AHF), a long-pass filter (HQ500lp, AHF), and
31 an avalanche photodiode detector (PDM series, MPD) were used to collect the entire
32 emission (400–800 nm) from the samples. Photon counting and exponential fitting of
33 the obtained PL decays were performed using the SymPhoTime software (version 5.3).
34 All Solid-State NMR (SSNMR) spectra were collected on Bruker Avance III wide-bore
35 NMR spectrometers at 600 MHz with a standard Bruker 4 mm double-resonance MAS

probe at room temperature. For 1D ^1H - ^{13}C CP-MAS experiments, the spectra of the samples were collected using a 2.5 μs $\pi/2$ -pulse, and 2048 scans, with a spinning frequency of 12 kHz. and the cross polarization/total suppression of spinning sidebands (CP-TOSS) pulse sequence was used to suppress the spinning sidebands. The 2D ^1H - ^{13}C HETCOR correlation spectra were collected using a 2.5 μs $\pi/2$ -pulse for ^1H and 4 μs for ^{13}C , and 512 scans, with a Lee-Goldburg/cross polarization (LG-CP) condition. All ^1H or ^{13}C chemical shifts are referenced to adamantane, with the ^1H signal of adamantane was calibrated to 4.6 ppm and the ^{13}C signal to 38.56 ppm.

2. Photocatalytic test

Reactions were carried in a Pyrex top-irradiation reaction vessel connected to a glass closed gas system. Water splitting reaction was performed by dispersing 20 mg of catalyst powders in an aqueous solution (50 mL) without any sacrificial agent. The reaction solution was evacuated several times to remove air completely prior to irradiation with a 300 W Xeon lamp with a working current of 15 A. The wavelength of the incident light was controlled by applying some appropriate long-pass cut-off filters. The detailed information of Xe lamp is shown in Figure S17. The temperature of the reaction solution was maintained at room temperature by a flow of cooling water during the reaction. The evolved gases were analyzed by gas chromatography equipped with a thermal conductive detector (TCD) and a 5Å molecular sieve column, using argon as the carrier gas (purchased from Beijing China Education AuLight Technology (CEAuLight) and Equipment Co., Ltd., GC7920). The AQE for H_2 evolution in the photocatalytic overall water splitting was measured under the illumination of a 300W Xe lamp (Beijing China Education AuLight Technology) with different bandpass filters ($\lambda=380\pm10$, 400 ± 10 , and $420\pm10\text{nm}$) used to provide the monochromatic light. The number of incident photons was determined using a fibre spectrometer (AvaSpec-ULS2048XL-EVO). Generally, for a single-component photocatalytic system, the AQE value was calculated according to equation by following process:

$$\alpha = \frac{\text{Number of evolved H}_2 \text{ molecules} \times 2}{\text{Number of incident photons}} \times 100\%$$

3. Calculation details.

The spin-polarised density functional theory (DFT) calculations were carried out using the Quantum ESPRESSO.^[1] The exchange–correlation functional was described by the Perdew, Burke, and Ernzerhof (PBE) parameterization of the generalized gradient approximation (GGA).^[2] The interactions between the electrons and the ions were treated within the projector augmented-wave (PAW) approximation.^[3] A cutoff energy of 450 eV was employed for the plane-wave basis set. A conjugate-gradient algorithm

was used to relax the atoms into their instantaneous ground state positions. The structural optimizations were not stopped until the atomic forces were less than 0.01 eV per Å. The first irreducible Brillouin zone was modelled based on the Gamma-centered scheme, where $4 \times 3 \times 1$ k-point grid was used in geometry optimizations and electronic structure analysis. The vdW-D3 method was adopted to describe the van der Waals interactions.^[4] A vacuum of 20 Å width was employed to avoid the interactions between periodic images in the z direction.

The change of the Gibbs free energy (ΔG) for each elementary step at the zero potential can be written as:

$$\Delta G(0) = \Delta E + \Delta E_{\text{ZPE}} - T\Delta S$$

where E is the energy that can be directly obtained from the DFT calculations. E_{ZPE} is the zero-point energy that can be calculated using $E_{\text{ZPE}} = 1/2 \sum h\nu$, in which ν is the vibrational frequency of a normal mode and h is the Planck constant. S is the entropy that can be calculated as:

$$S(T) = k_B \sum_i \left(\frac{h\nu_i}{k_B T} \frac{1}{\exp\left(\frac{h\nu_i}{k_B T} - 1\right)} - \ln \left(1 - \exp\left(\frac{h\nu_i}{k_B T} - 1\right) \right) \right)$$

k_B and ν_i are the Boltzmann constant and vibrational frequency, respectively.

Table S1. The S_{BET} , pore volume, band gap value, time-resolved PL spectra (τ_{avg}) and RhB removal time of g-C₃N₄ samples treated for different time.

Sample	C ₃ N ₄ yields	S_{BET} (m ² /g)	Pore volume(cm ³ /g)	Band gap (eV)	τ_{avg} (ns)	co-catalyst mass ratio	RhB dark adsorption/ removal time
Pt/PtCo@CN	51.21%	44.78	0.27	2.52	5.72	0.833% Pt/ 0.094% Co	75% / 120 min
Co _{SAS} /PtCo@CNN	8.32%	277.86	1.61	2.86	6.92	0.495% Pt/ 0.128% Co	43% / 5 min

1 **Table S2.** EXAFS fitting parameters at the Pt L₃-edge for various samples ($S_0^2=0.86$).

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3 ^a*N*: coordination numbers; ^b*R*: bond distance; ^c σ^2 : Debye-Waller factors; ^d ΔE_0 : the inner potential

	shell	CN	R(Å)	σ^2	ΔE_0	R factor
Pt foil	Pt-Pt	12	2.76±0.01	0.0048	7.7±0.5	0.0028
sample	Pt-N	1.8±0.1	2.01±0.01	0.0066	6.8±0.9	0.0035
	Pt-Co	1.2±0.2	2.66±0.01	0.0075		

4 correction. *R* factor: goodness of fit.

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Table S3. EXAFS fitting parameters at the Co K-edge for various samples ($S_0^2=0.74$).

^a*N*: coordination numbers; ^b*R*: bond distance; ^c σ^2 : Debye-Waller factors; ^d ΔE_0 : the inner potential correction. *R* factor: goodness of fit.

	shell	CN	R(Å)	σ^2	ΔE_0	R factor
Co foil	Co-Co	12	2.49±0.01	0.0063	6.8±0.4	0.0017
sample	Co-C	0.8±0.2	1.60±0.03	0.0059±0.0021	5.3±2.9	0.0099
	Co-N	3.8±0.9	2.13±0.02	0.0059±0.0021		
	Co-Pt	2.1±0.9	2.70±0.02	0.0046±0.0026		

Table S4. Comparison of photocatalytic activity of overall water splitting over C₃N₄-based materials

Materials	Reaction conditions	Cocatalyst	Incident light	H ₂ rate (μmol g ⁻¹ h ⁻¹)	O ₂ rate (μmol g ⁻¹ h ⁻¹)	Activity per active site (μmol mg ⁻¹ h ⁻¹)	Ref
CoSAs/PtCo @CNN	300 W Xe lamp, 20 mg	0.5% Pt, 0.13% Co	λ > 300 nm λ > 400 nm	300.9 85.7	65.6 21.3	48.3 13.8	This work
3D g-C ₃ N ₄ NS	300 W Xe lamp, 50 mg	1 wt.% Pt, 3 wt.% IrO ₂	λ > 420 nm	102	50	2.6	5
Pt/g-C ₃ N ₄	300 W Xe lamp, 200 mg	3 wt.% Pt, 1 wt.% CoO _x	λ > 300 nm λ > 420 nm	61 6	31.5 3	1.5 0.15	6
Co ₃ O ₄ /HCN S/Pt	300 W Xe lamp, 20 mg	1 wt.% Pt, 3 wt.% Co ₃ O ₄	λ > 300 nm	175	85	4.4	7
Pt/CoP/g-C ₃ N ₄	300 W Xe lamp, 80 mg, pH = 3	3 wt.% Pt, 3 wt.% CoP	λ > 300 nm λ > 400 nm	262.5 26.3	125 12.5	4.4 0.4	8
Z-scheme α-Fe ₂ O ₃ /2D-C ₃ N ₄	300 W Xe lamp, 10 mg	3 wt.% Pt, 0.1 wt.% RuO ₂	λ > 400 nm	38	19	1.2	9
Na-CN	300 W Xe lamp, 100 mg	1 wt.% Pt	λ > 420 nm	315	152	31.5	10
(C _{ring})-C ₃ N ₄	300 W Xe lamp, 30 mg	3 wt.% of Pt	λ > 300 nm λ > 420 nm	369.6 149.8	183.1 73.2	12.3 5.0	11
PCN/LaOCl	300 W Xe lamp, 50 mg	0.5 wt.% Pt, 0.2 wt.% CoO _x	λ > 300 nm λ > 400 nm	446 160	214 76	63.7 22.9	12
CoO/g-C ₃ N ₄	LED, 50 mg	30% CoO	λ > 400 nm	50	28	0.17	13
C ₃ N ₄ /MnO ₂	300 W Xe lamp, 100 mg	N/A	λ > 420 nm	55	28		14
g-C ₃ N ₄ -carbon dots	300 W Xe lamp, 100 mg	N/A	λ > 420 nm	5	2.5		15
Z-scheme MnO ₂ /g-C ₃ N ₄	300 W Xe lamp, 100 mg	3 wt.% of Pt	λ > 400 nm	61	29	2.0	16
CdS/Ni ₂ P/g-C ₃ N ₄	300 W Xe lamp, 50 mg	N/A	λ > 420 nm	15.6	7.75		17
CoNiP-PCN	300 W Xe lamp, 10mg	N/A	λ > 300 nm	239.3	Trace		18
WC _{1-x} /g-C ₃ N _x	300 W Xe lamp, 40 mg	N/A	λ > 300 nm	84	41.7		19

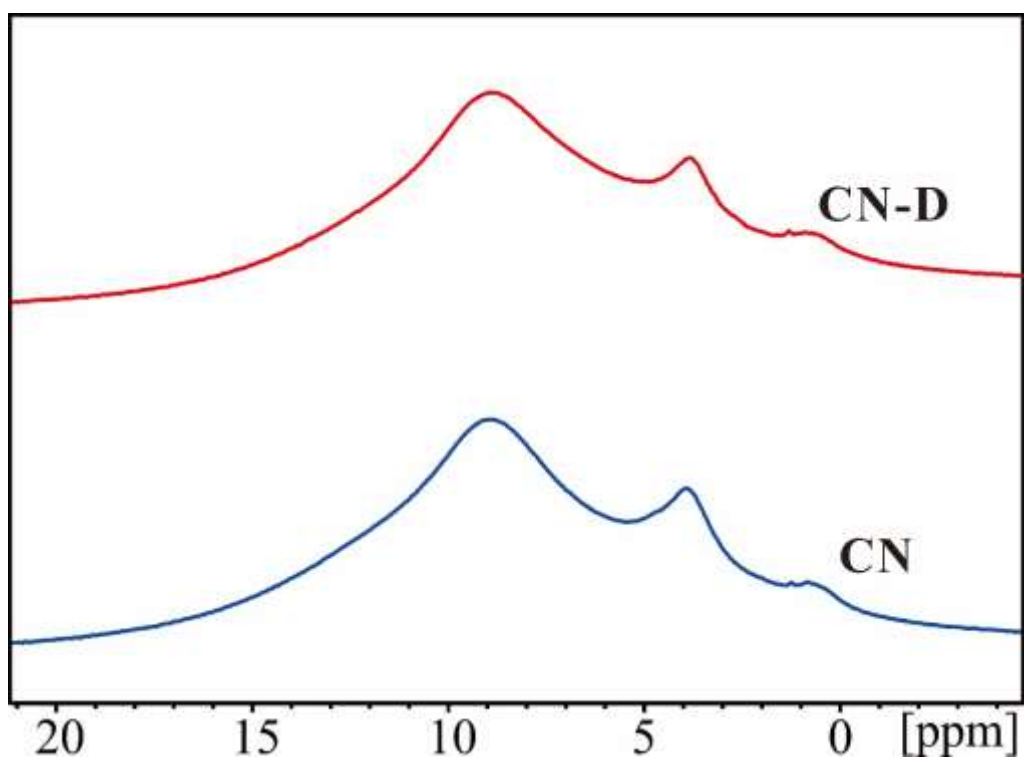


Figure S1. One-dimensional (1D) ^1H magic angle spinning (MAS) NMR spectra of the bulk graphitic carbon nitride (CN) before and after D_2O treatment, where the D_2O -treated CN is denoted as CN-D.

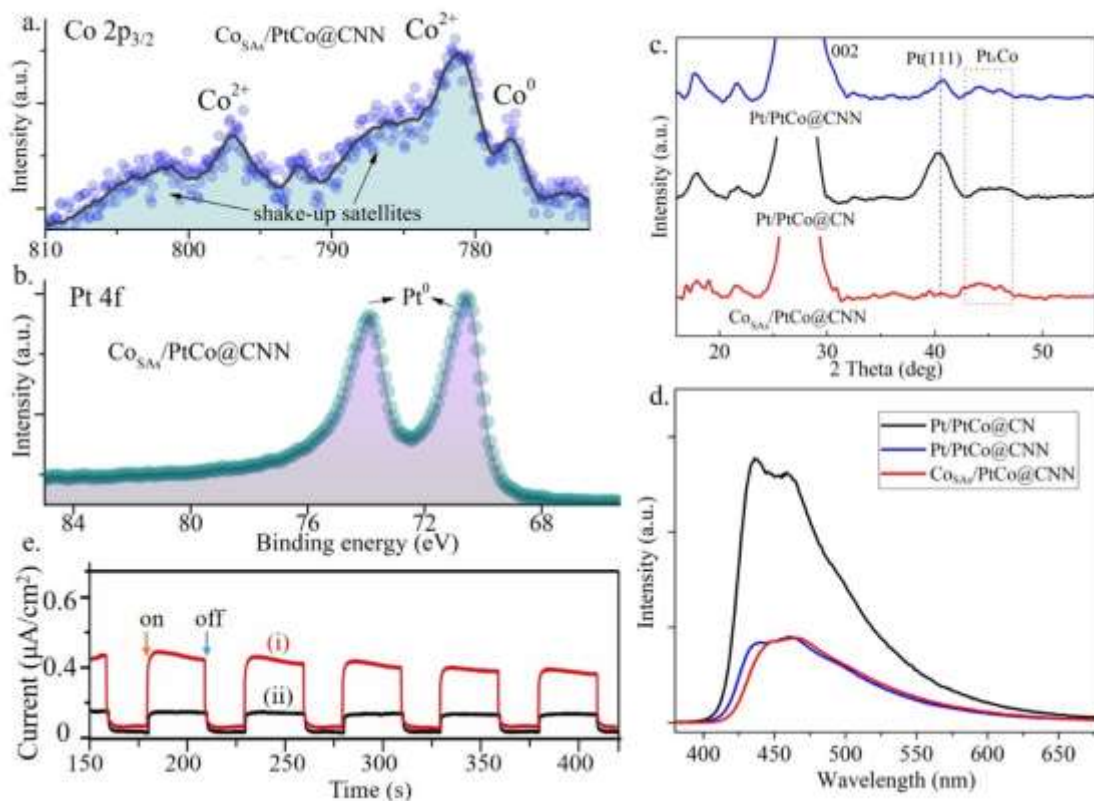


Figure S2. CoSAs/PtCo@CNN electronic structures. XPS spectra of CoSAs/PtCo@CNN sample taken at (a) Co 2p_{3/2}, (b) Pt 4f. (c) XRD images and (d) steady-state PL emission spectra of CoSAs/PtCo@CNN and referenced Pt/PtCo@CNN, Pt/PtCo@CN samples. (e) Transient photocurrent density curves of CoSAs/PtCo@CNN catalyst (i) and Pt/PtCo@CN catalyst (ii) at 0.2 V vs. Ag/AgCl in 0.5 M Na₂SO₄ aqueous solution under sequential illumination condition with an interval of 20 s on/off switch.

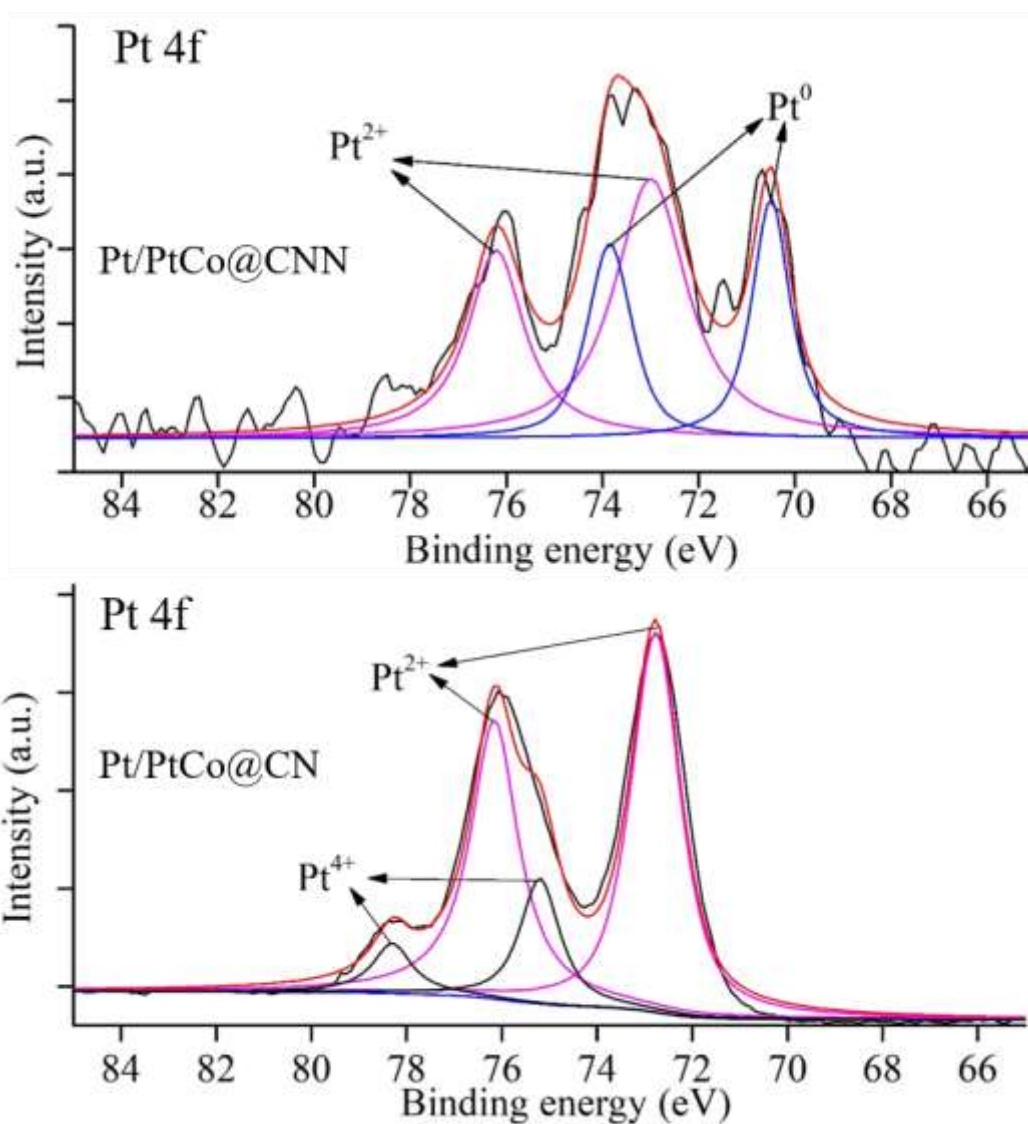
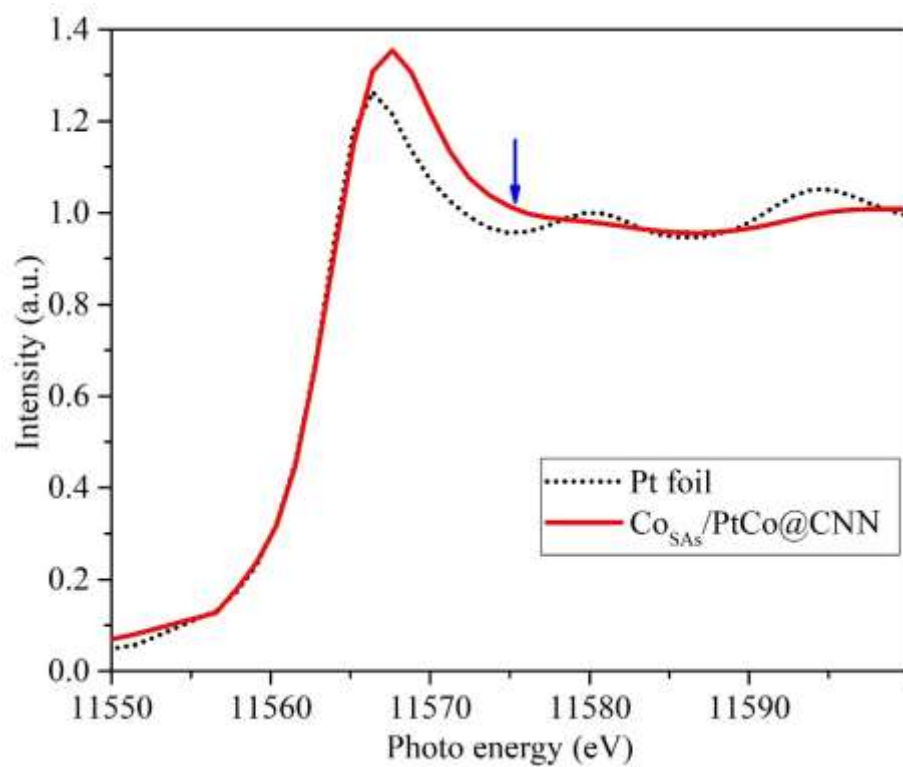


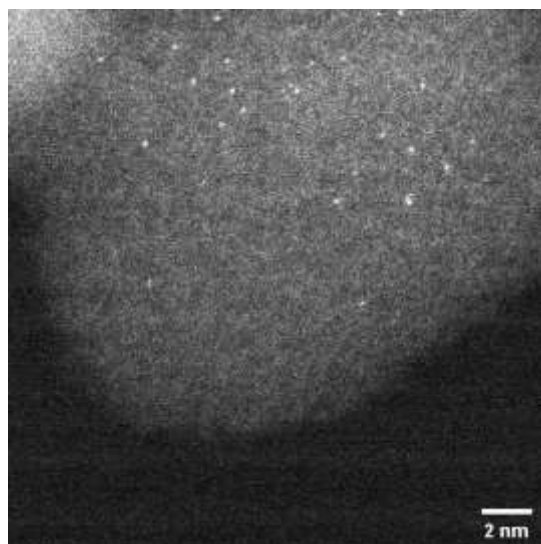
Figure S3. XPS spectra of Pt/PtCo@CN and Pt/PtCo@CNN sample taken at Pt 4f.



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2 Figure S4. XANES spectra taken at Pt L₃-edge with metal foils as references.

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Figure S5. HAADF-STEM image of the CoSAs/PtCo@CNN catalyst.

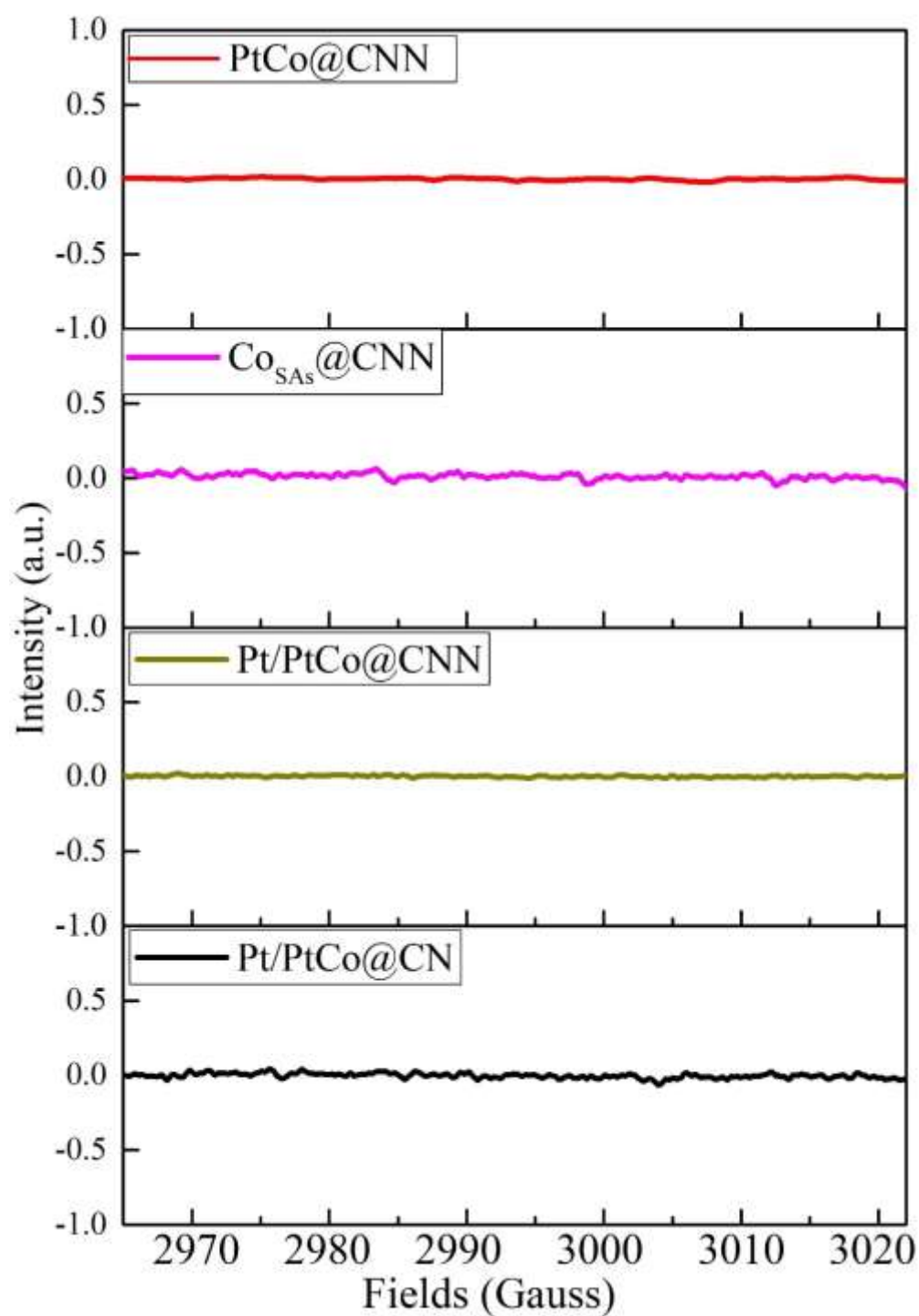
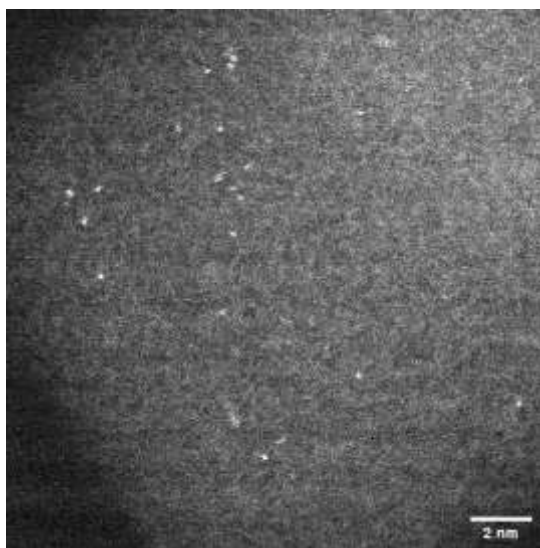


Figure S6. In-situ EPR spectra of the DMPO–OH spin adduct obtained in Pt/PtCo@CN, Pt/PtCo@CNN, CoSAs@CNN, and PtCo@CNN suspensions under same UV-vis irradiation.



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2 **Figure S7.** HAADF-STEM image of the CoSAs@CNN catalyst.

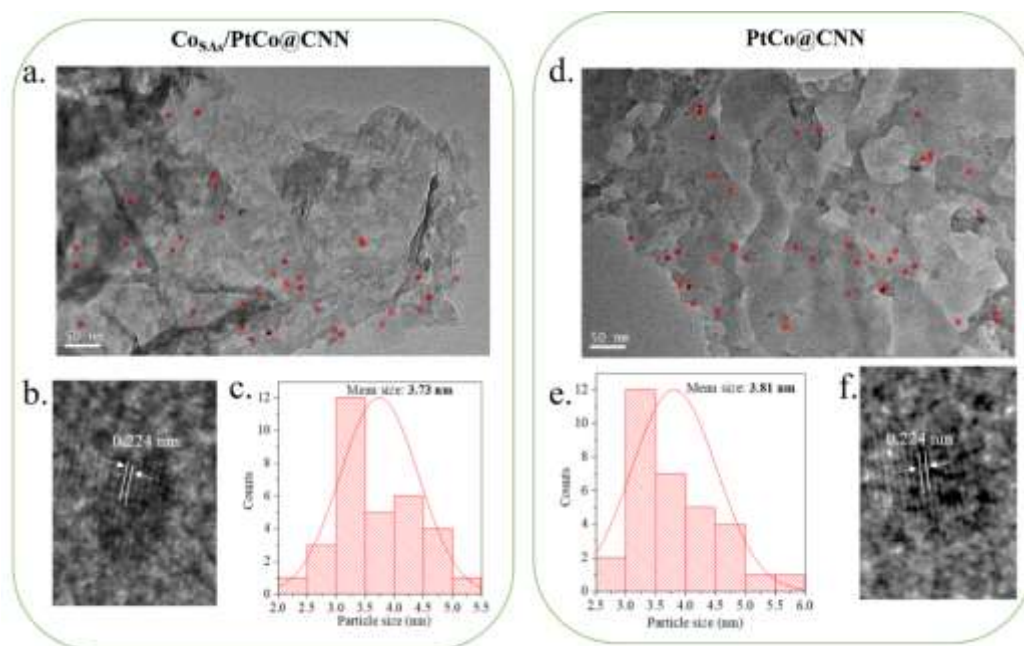
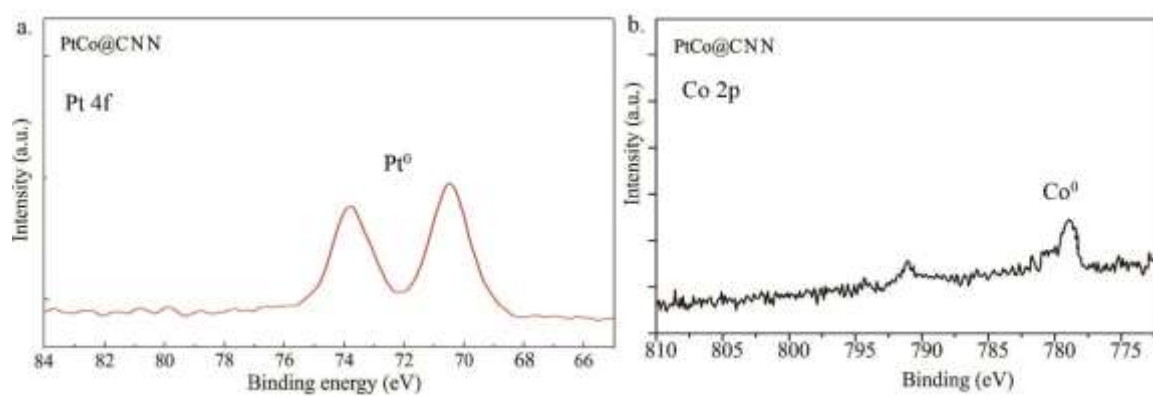


Figure S8. (a). TEM image, (b). related crystal plane and (c). extracted particle size distributions for the CoSAs/PtCo@CNN sample; (d). TEM images, (e). related crystal planes and f. extracted particle size distributions for the PtCo@CNN sample.

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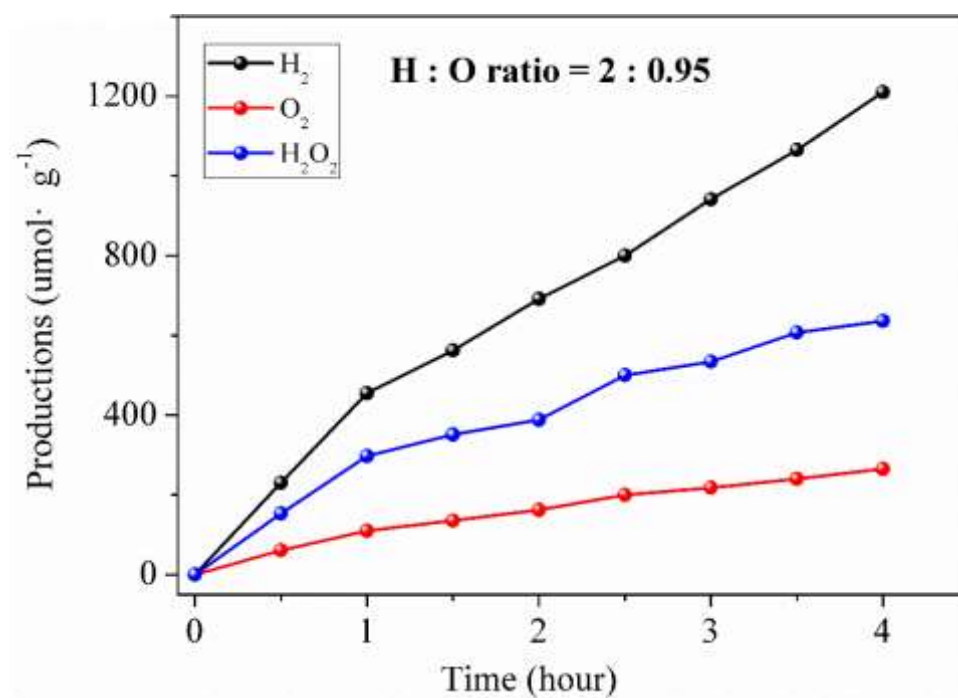
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Figure S9. XPS spectra of PtCo@CNN sample taken at (a) Pt 4f, (b) Co 2p.

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3 **Figure S10.** Photocatalytic measurements for overall water splitting over
 4 CoSAs/PtCo@CNN catalyst. Reaction conditions: 20 mg photocatalyst and 50 mL pure
 5 water were illuminated under UV-visible light irradiation ($\lambda > 300$ nm).

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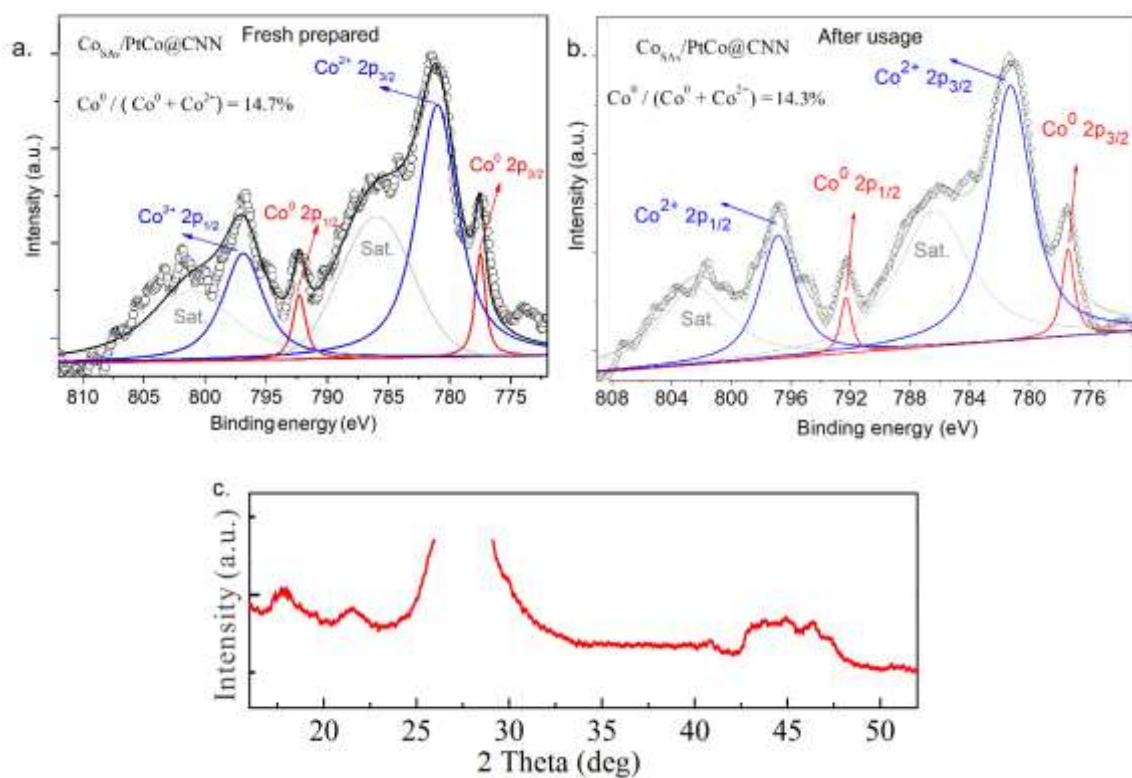


Figure S11. XPS spectra of CoSAs/PtCo@ CNN sample taken at (a) fresh prepared state and (b) 3-cycling usage state at Co 2p. (c). XRD spectrum of CoSAs/PtCo@CNN sample taken after 3-cycling usage.

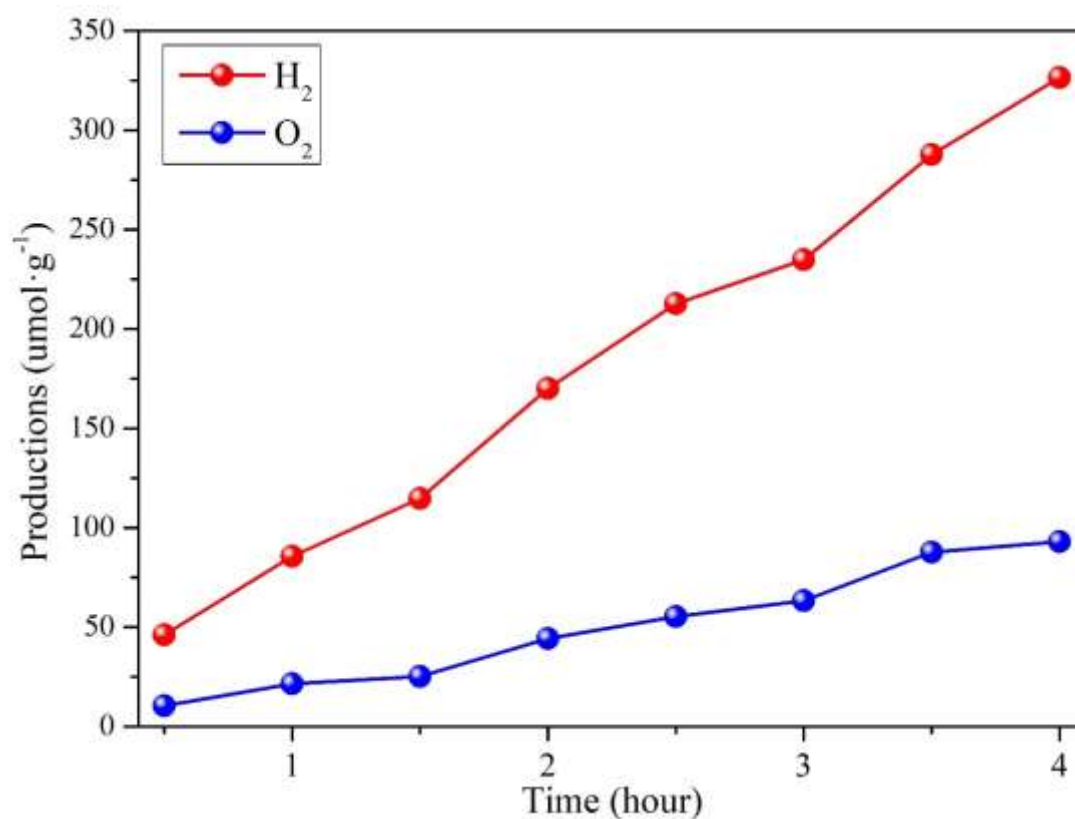


Figure S12. The photocatalytic overall water splitting activities over CoSAs/PtCo@CNN sample under visible light irradiation (>400 nm).

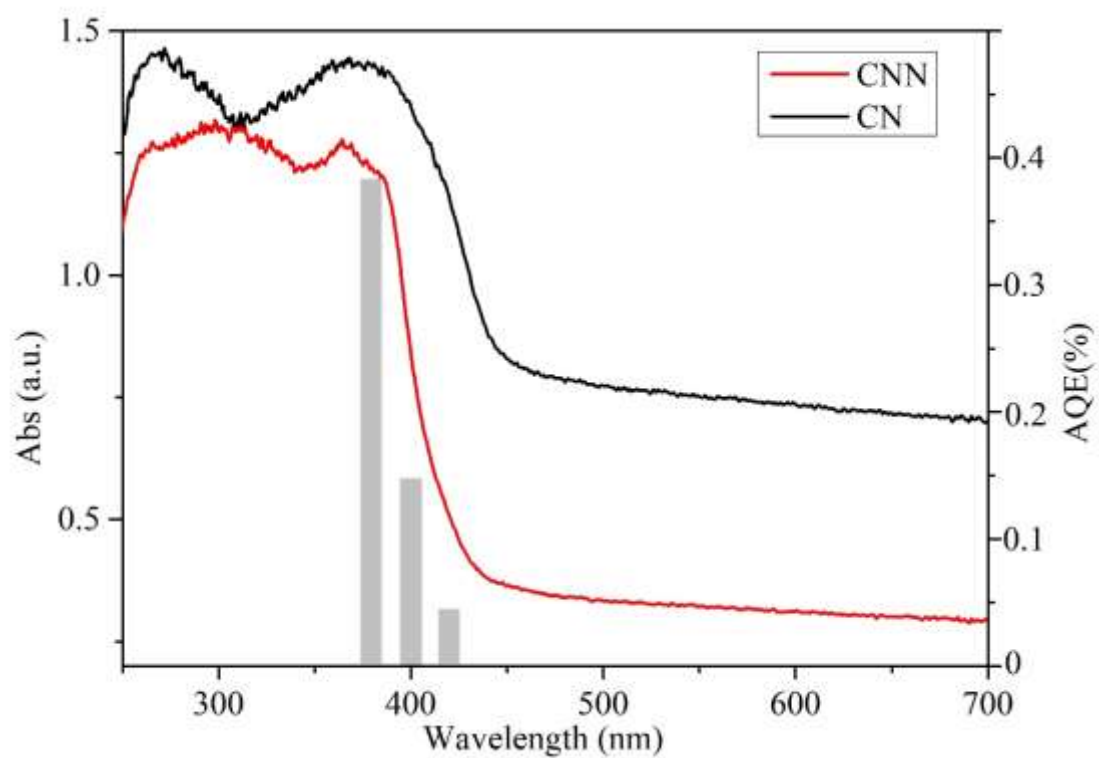


Figure S13. The wavelength-dependent apparent quantum efficiency (AQE) of overall water splitting over CosAs/PtCo@CNN sample.

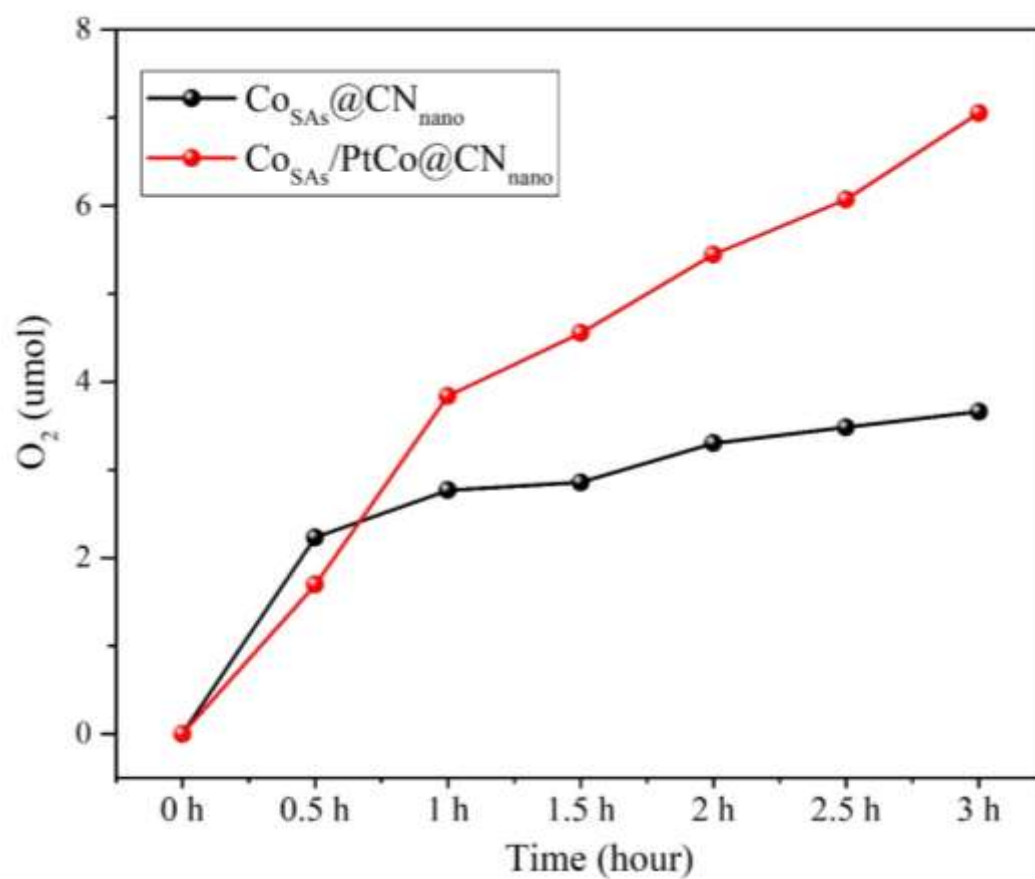


Figure S14. The photocatalytic OER activities over $\text{Co}_{\text{SAs}}/\text{PtCo}@\text{CNN}$ sample under UV-visible light irradiation

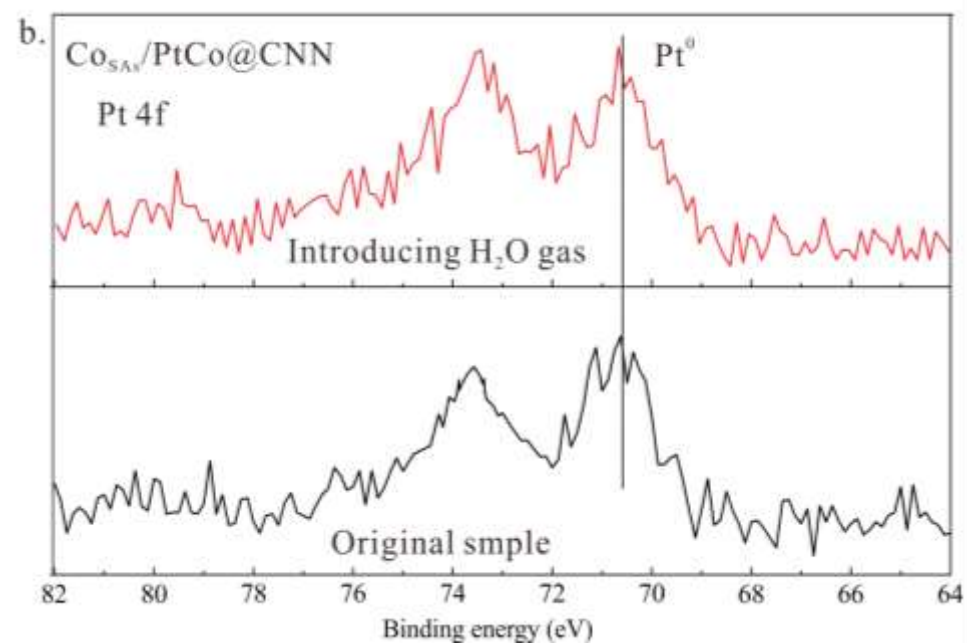
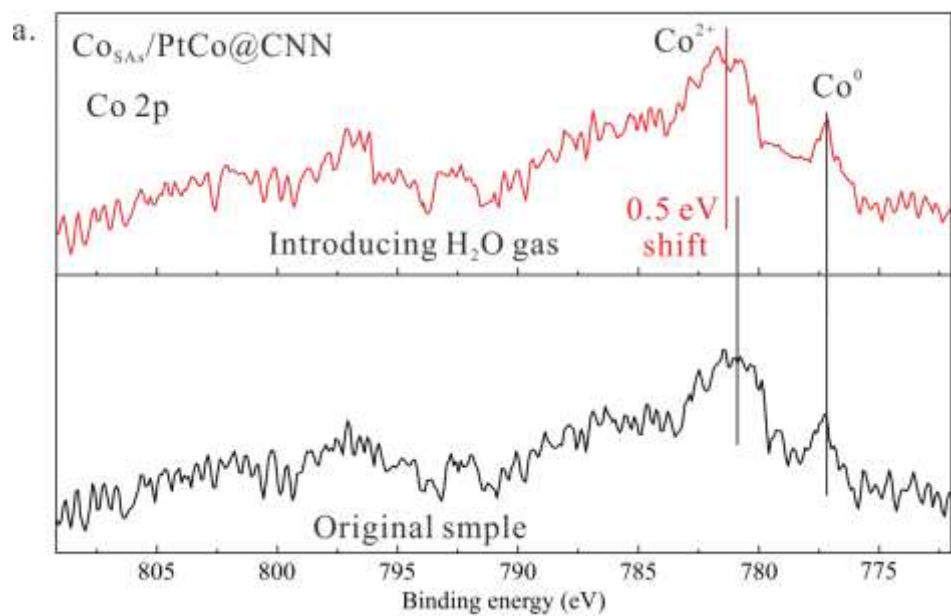
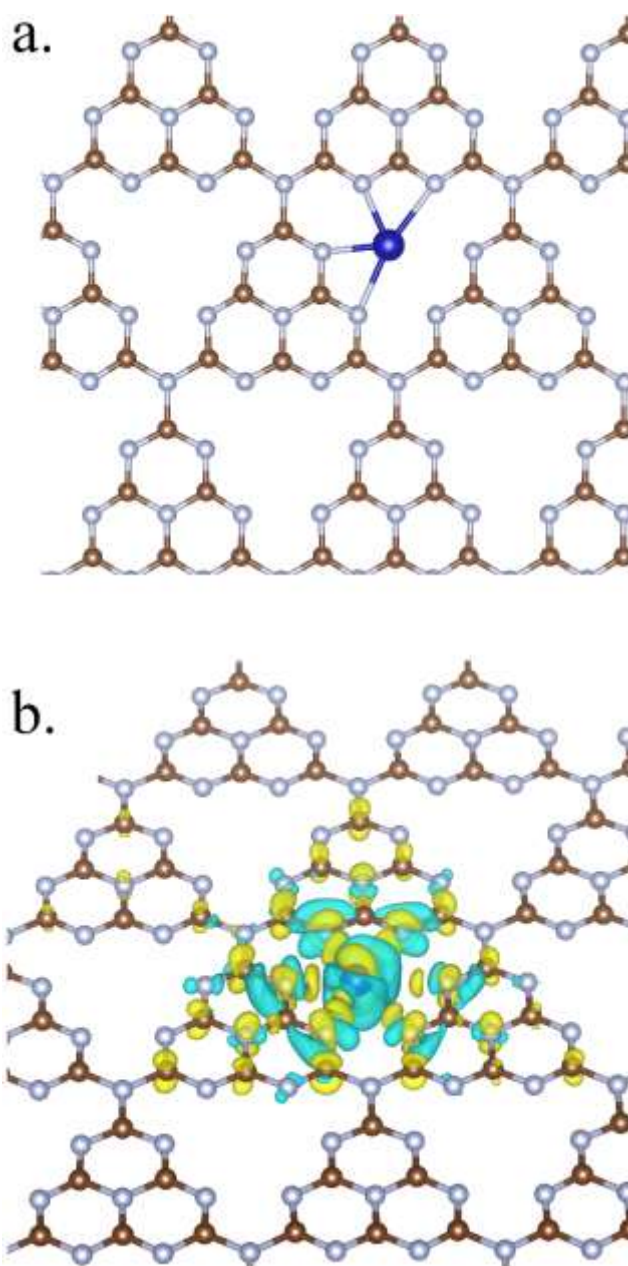


Figure S15. Near-atmospheric X-ray photoelectron spectroscopy (NAP-XPS) recorded before and after introducing a 10 mbar H_2O gas purge taken at (a) Co 2p and (b) Pt 4f.

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3 **Figure S16.** (a). Calculated CoSAs coordination environment and (b). charge density
4 differences for CoSAs/PtCo@CNN catalyst. Yellow and cyan areas represent the net
5 charge density is positive and negative, respectively.

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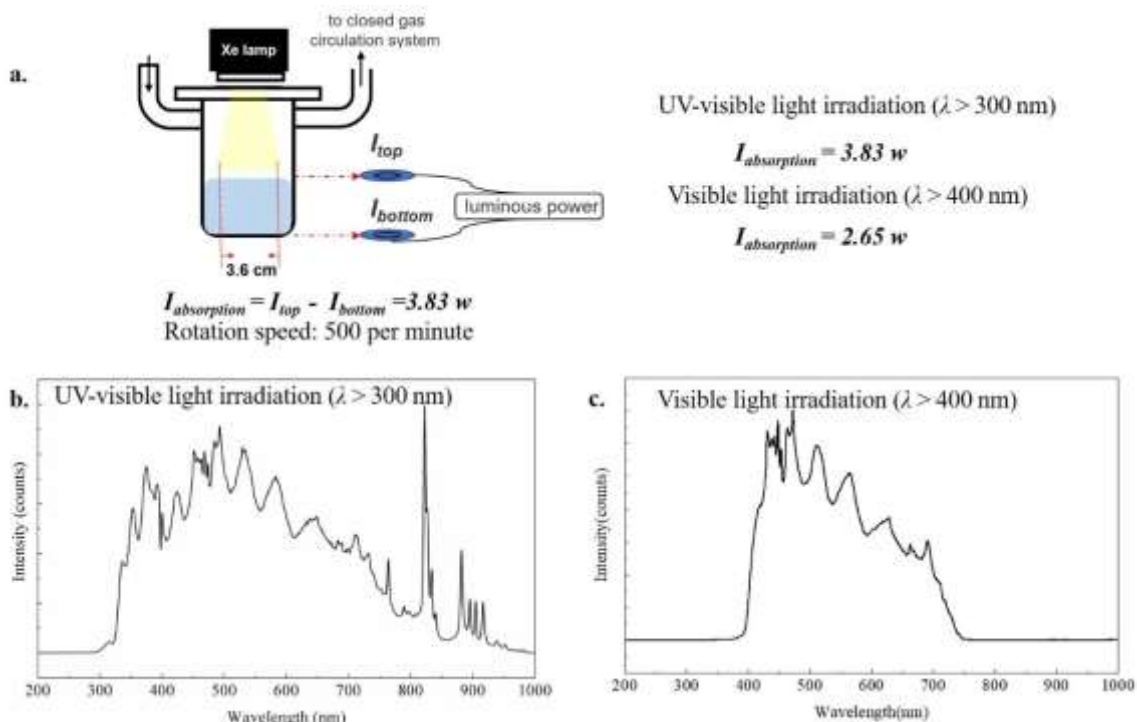


Figure S17. (a). Schematic layout of photoreduction reactor and light intensity. (b). Wavelength distribution of each band in the process of Xenon lamp photoreduction ($\lambda > 300 \text{ nm}$). (c). Wavelength distribution of each band in the visible light irradiation ($\lambda > 400 \text{ nm}$).

As shown in Figure S16a, the irradiation diameter of 300 W Xenon lamp is 3.6 cm, the accepted light intensity under the full-band wavelength is 3.83 W, and the specific band distribution is shown in Figure R6b. For the visible light experiment, we use the 400 nm long-pass cut-off filter, which is often used in visible-light photocatalysis. The specific band distribution is shown in Figure R6c, and the light intensity received under visible-light is 2.65 W.

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